

APTS Logic 7 101

Instructor Guide

(v1.3)

Introduction

Welcome to the Instructor Guide for the Apple Pro Training Series Level One Logic 7 class. This guide is designed as a tool to help give you ideas about how to teach your class, and get the most from the Logic book. The information in this guide is merely a group of suggestions that you are in no way obligated to implement, but hopefully you find helpful to you and your students.

Main Goal!

Obviously anyone attending your class wants to learn how to use Logic. It is also important to consider that those attending your class may also want to become an Apple Certified Logic User. In order to do that they must take a test which is derived from the Logic book that you are teaching from. As the test is derived from all areas of the book, it is imperative that all lessons in the book are sufficiently covered.

Main Challenge

Simply put the main challenge is... **TIME!** Everyone's classes are organized a little differently, there are different numbers of students, and instructors have different teaching styles. You must make it a priority to create a system for yourself that will allow you the time to cover the entire book, otherwise you may fall short of the main goal.

Instructor Guide Layout

The instructor guide is organized into three types of headings which reflect the same structure as the book.

Section Headings: The book consists of four main sections, which are over-viewed on page 2 of the book.

Lesson Headings: It is important for a student to know what they are about to learn, and why they should learn it. The book introduces each lesson with great metaphors and practical examples of the topics that are going to be discussed. Each lesson needs a strong introduction from you, but don't feel you have to read these word for word. Study these introductions to get some ideas, but interject your own thoughts and experiences and make them into your own!

Each lesson ends with a section called "What You've Learned" which gives a point by point overview of the topics covered. At the end of each lesson, recap each of these points, and confirm that everyone in the class is comfortable with the material.

Giving your class a quick verbal quiz about the lessons content help you as well as your students know if there are any topics that need to be reviewed.

Exercise Headings: Within each lesson are exercise headings. In the book the main exercise headings are indicated in blue text, and supporting exercise headings are indicated in bold black text. In this guide main exercise headings are shown in bold underlined text, and supporting exercise headings are not referenced. Like the lessons, these exercise headings need their own mini introductions with your personal touch!

Page Numbers: The numbers below headings indicate page numbers from the Logic book. Page numbers are then followed by page comments. Some section headings have a page number, but no following comment. This is intentional to serve as a place holder. Feel free to add your own comments in these sections.

Page Comments (The Three T's)

The Three T's represent the three categories that page comments fall into. They are color coded so that in a classroom environment they quickly convey the type of information they are intended to give.

TS: Teaching Strategies (or Teaching Suggestions) are written in blue. These give you helpful hints on how to approach teaching a particular concept, and how maximize your use of time.

TT: Teacher Tips are written in green. These are not tips intended for the teacher but instead represent information that you the teacher can give to the students. This is information that is not discussed in the book, but may be of interest to your class.

TC: Technical-Concerns are written in red. They provide an awareness of potential technical issues that could pose a problem when teaching your class. This includes compatibility issues, software version changes, or incorrect information in the book.

Modes of Class Participation

Throughout this guide there will be references to the *modes of class participation* discussed in the Train the Trainer class you attended. Here is a re-cap of those modes.

Holding-Hands: Have students follow you on their computers one step at a time. This is useful when they are first introduced to a new section of the software. It is also very helpful in lessons that can create changes that are difficult to reverse. It is imperative to emphasize that students should not do anything on their computers until asked to, otherwise you will be running around the classroom trying to fix their mistakes.

Show-&-Go: The students watch you move through the steps of an exercise, and then when completed, they are given time to work through the exercise themselves.

The “Show” part gives students a chance to focus on what you are saying, without the distraction of trying to follow along on their computer. It is always very important to announce that what you are about to show them is something they are going to have to do! When they know they have to do it, it makes them much more attentive. The “Go” part allows students to move through the exercise steps at their own pace, but be sure to state how much time will be allocated to the exercise. While students work through the steps, this allows you to move about the classroom, and help students on an individual level if they have problems.

Demonstration: You simply show how the software works, without students participation. This is usually the least preferred method, as it looks more like a sales presentation than an interactive class. It’s a well suited option when responding to student questions, and can be a big time saver if you are running into time constraints.

Use of Terms

It is very important to use proper terminology when discussing Logic with your class. Using incorrect terms leads to confusion, and potentially causes students to not understand certification exam questions.

Key Commands

The book often indicates key commands that are available. It is very important to emphasize that most all key commands in Logic can be re-programmed, therefore one should always know where a feature can be accessed if the expected key command is not available.

Participate in this Guide

This guide is meant to be an evolving tool for everyone certified to teach Logic. You will likely discover your own methods for teaching Logic, and if you would like to submit your discoveries and ideas to be included in this guide please email them to brock@cras.org.

Section I

Logic Overview

7

TS: The first three lessons are all about getting comfortable with how Logic looks and feels. Many students in your class will already have experience with either Logic, or other programs that are similar to Logic. Survey your class to see what they are experienced with. As you have cover how Logic works, when appropriate point out when things might be similar to other programs.

Lesson 1

Exploring the Workspace

9

TS: Lesson 1 covers a lot of ground. It gives immediate satisfaction by taking a student through Logic from opening a song, to bouncing a final mix. On the whole, you can look at this lesson as a brief introduction, but there are critical concepts that are only covered in this lesson.

TS: The Holding-Hands approach is a good way to start student participation, and give them confidence in using Logic. Be sure to emphasize that they should not do anything on their computers unless you specifically tell them to.

Launching Logic

10

TC: The book never mentions the Startup Assistant. If a student uses the demo version of Logic 7 Express included with the book, the first time he opens Logic, they will be guided through the setup process. A concern is that if you mention the Startup Assistant, then you can get distracted from the main scope of the lesson. A good place to later talk about the Startup Assistant would be as an add-on to the environment Lessons 11 and 12.

TS: Show Package Contents at the beginning of this exercise is not critical to use of the program, and may lead to a lot of questions that do not pertain to the main mission of the exercise. You may want to skip it all together and start at step 1.

Opening a Song File

13

Getting to Know the Arrange Window

15-18

TC: These pages give an overview of the Arrange window. Many of these features are addressed in greater detail later in the book, so resist the temptation to spend more than just a minute or two in this area.

19

TT: The book says the secondary tool is also referred to as the alternative tool. The manual only refers to it as the alternative tool.

20

TS: The Exploring the Transport Panel section can better be thought of as the beginning of the Moving the Song Position Line section that starts on page 22. Introduce this heading as you would any of the other sections within a lesson.

TC: You don't want students to have the impression that the Transport Panel in the Arrange window will always appear. This is a good place to show how the Local Menu's View heading gives various options for customizing the Arrange window.

21

TC: Controlling Playback Using the Number Pad: Consider skipping this section if you are using laptop. Although it is possible to put a laptop into a numeric keypad mode, this is likely to become distraction. Many people use USB numeric keypads with their laptops when using Logic.

Moving the Song Position Line

22

TS: Skip the note at the top of the page, as cycle mode is covered later in the book.

27

TS: The note about using the control key for finer resolution potentially distracts from the main objective. Using control is discussed later in the book.

28

TT: On step 2, since the position slider involves the area between the song start and end points, this is a good place to introduce students on how to change the song start and end points.

Setting a Tempo

29

About the Arrange Window Channel Strip

30

TS: This and the next four sections represent a major new direction in this Lesson, as these topics relate to manipulating and generating sound within Logic. This is a great place to relate how Logic mirrors a studio filled with mixers, outboard gear, and instruments. This metaphor sets a point of reference that you can continually go back to throughout the book.

32

TT: Channel Strip Only has a key command available, which is very nice for users with small screen sizes. It would be good to mention the key command, but defer setting it until customizing key commands is discussed in the book.

Using DSP Plug-ins

33

Using Audio Instruments

42

TC: The last paragraph of this page indicates that Logic supports 192kHz sample rates. This is true for Logic 7 Pro, but not Logic 7 Express.

44

TT: The Caps Lock Keyboard is shown on this page. Often students inadvertently open this window, and since it has no Close button, they can't figure out how to get rid of it. Emphasize that the Caps Lock key is the only way to turn on or off the Caps Lock Keyboard.

Using Channel Strip Settings

50

Bouncing a Song

54

TS: This is the only place in the book that bouncing is addressed, but you may want to consider holding off on this topic until after Lesson 10 is completed. This topic can become a huge distraction as lot of technical concerns need to addressed, but are better suited after the student has a better understanding of Logic.

TT: It is worth noting that you can select more than one bounce destination at a time. Very cool to burn a disk, and create a .mp3 at the same time!

What You've Learned

60

TS: This lesson ends with bouncing a file, but the student has not been told to save the lesson. This is a good time cover the importance of saving files as they work. "Save As..." should also be introduced at this time.

Lesson 2

Exploring the Editing Windows

63

TS: In addition to learning about various editing windows, a major component of this lesson is window management. Many programs like Logic have started to consolidate the use of many windows into a few multi-functional windows. Assure students ahead of time that although Logic has a lot of different types of windows, in this Lesson they will learn elegant, and flexible solutions to managing Logic's massive power!

TS: At this stage, Show-&-Go tends to work best for working through these exercises. If at the end of the lesson you find that you are ahead of schedule, then give the students some time to play with the things they've just learned.

Getting to know the Editing Windows

64

TS: This section is simply a quick introduction to the various windows in Logic. Although the Lesson introduction says to work carefully through every step, You can quickly guide the students through the windows in this section using the Holding-Hands technique. Don't feel like you have to follow each step exactly. As long as you introduce each window, and say a few words about each one, you've accomplished the goal.

TS: Remember that most of the windows will be covered in greater detail in later lessons. Don't spend more than a few minutes on each window. You will likely get a barrage of questions like "Can it do this?". It is okay to say "We'll be answering those questions in a later lesson."

65

TC: The first paragraph indicates that the Track Mixer add a track in the Arrange window, that it automatically appears in the Track Mixer. There are several exceptions to this, such as when multiple tracks are assigned to the same output. Also, if the Track Mixer is in Global mode, the Track Mixer reflects the Environment instead of the Arrange window. This is not something that needs to be stated to your class unless someone asks.

67

TT: This is the only time in the book that the Score editor is mentioned. If you have students who are interested in learning more about the Score editor, there is an excellent introductory lesson in the Advanced Logic Pro 7 book by David Dvorin that is part of the same Apple Pro Training Series.

69

TC: In the previous exercise, the student was told to make sure that no regions were selected in the Arrange window. Before the Hyper Editor can be opened, they must select the Launch House Pad region.

71

TS: This is the first mention of the Environment in Logic. At this stage in the book, it is not necessary to get into the Environment, and glancing over it can end up creating a lot of confusion. Lessons 11 and 12 allocate an appropriate amount of time to tackle the Environment concept. As the Environment is not really an editor in the same sense as the other windows being discussed in this lesson, it would be acceptable to skip covering the environment at this time.

75

TT: Advise students to be very careful when using the Sample editor, as it can destructively re-write the audio files they are working with.

Working with Windows

77

TT: The Making Windows Active section has the students opening a few different windows. Use this as an opportunity to show that it is possible to open multiple windows of the same type (ie. 2 different Matrix windows). Students often bury a window, and instead of finding it, they open a second window of the same type without realizing it. Using the technique shown in this section can help them find their lost windows.

79

TT: Another easy way to identify a floating window is that they don't have a minimize, or maximize button.

TT: It should be mentioned that key commands do not apply to floating windows.

TC: The note at the bottom of the page indicating floating windows disappear when switching to another application is no longer true for the Transport window when using Logic 7.1 on OS 10.4.2.

Linking Windows

81

TS: Some people have a difficulty understanding the difference between Link, and Content Link. If they don't seem to get it within a few minutes of working through the exercise, offer to help them individually at a later point. Don't spend more than 5 minutes on this topic.

Using Screensets

84

TS: It cannot be emphasized enough how important screensets are in Logic. This is what makes Logic's massive window set manageable. Logic is the program that pioneered this concept in audio programs. Imagine if every type of program had screensets! Imagine if you had screensets for your life! Just push a button and everything you need to do what you want to do is there!!!

What You've Learned

90

Lesson 3

Understanding Workflow Techniques

93

Creating a New Song

94

TT: The second paragraph indicates that Song Templates contain no media. It is possible to save a song template, and any media within the template will be there when you use the template again. This can be very useful in commercial applications when the same music bed might be used over and over again for the same client.

101

TT: The Creating a Project Folder section the student just completed had the student select the copy all samples, instruments, impulses checkboxes. The potential problem is that if the user is using large sample libraries (such as the Vienna Symphonic Library), massive amounts of data will be copied every time the load a sound. This will eventually cause the users hard drive to fill up. Typically, supporting files for a song only need to be consolidated into one space when moving the file for archiving or moving it to another computer. Under File in the main menu, the Save as Project option gives the same consolidation features, but can be used when saving the file.

TC: Sample rates above 96kHz are not possible in Logic Express.

TS: When discussing sample rates, its very common to also discuss bit resolution. Setting the bit resolution will be discussed later in Lesson 8 Recording on page 330.

Importing Audio into the Arrange Window

102

TT: In a bit of irony, one thing you cannot import from the File > Import option is audio files.

106

TC: With Logic 7.1, it is not necessary to do steps 13 and 14.

Importing MIDI into the Arrange Window

107

TT: When importing Type 0 (multiple tracks combined into one track) standard MIDI files, Logic automatically creates a separate track for each MIDI channel.

Zooming and Magnifying

109

TT: The note indicates that zooming can be done with the Ctrl-Arrow key commands. Keep in mind that this may be programmed differently on some systems. Interestingly, holding Control to switch to the Magnifying Glass cannot be changed.

Selecting with the Arrow Tool

113

TS: Be careful to use proper terminology. Many like to refer to this as the pointer, but when its used within an editor, it should be called the Arrow Tool.

Rubber-Band Selecting

114

Using the Cycle Mode

116

Using the Skip Cycle Mode

120

Dividing Regions

121

123

TS: When dividing MIDI regions, you may want to diagram on a white board what the Keep, Split and Divide options do.

127

TC: Step 3 has the student use the Split Regions command. Technically the operation is a "Divide" operation. Using many different terms for the same thing, causes confusion. Obviously you can't change it when Logic itself does this, but you can point this out to clarify what is really being done.

Moving and Copying Regions

128

Naming Regions

130

Hiding Tracks (Logic Pro Only)

133

Using the Catch Function

135

Using Scroll in Play

137

Using Key Commands

138

TT: Emphasize the importance of not resetting someone else's key commands without saving them first.

What You've Learned

144

Section II

Creating a Song

145

TS: Section II includes lessons 4 through 10. It takes the student through building a song from start to finish. This is the main reason your students have come to you! Emphasize to your class that this is where the real fun starts!!

Lesson 4

Editing Audio Regions

147

Exploring the Audio Window

148

TC: The first note on this page indicates that the Sample Editor is Logic's only destructive editor. There are operations in the the Audio window that can also be destructively applied.

151

TC: The note indicates that Cmd-9 will open the audio window, but remember that this key command can be changed.

154

TC: When importing .mp3 files, there have been sporadic problems with certain combinations of Logic and Quicktime. If you encounter problems when importing files in this exercise, make sure your versions of Logic and Quicktime are up to date.

156

TS: The Selecting Audio Files vs. Audio Regions section needs to be emphasized. This is definitely a topic that you want to take your time with, and make sure everyone understands before moving on.

Using the Sample Editor

157

TC: The note on this page indicates that the Arrange window is also a sample accurate editor. This is not true.

TS: The Sample editor does not ask if you'd like to make a backup before doing a destructive process (as it did in older versions). You may want to consider using the Holding-Hands approach to avoid problems.

Auditioning Audio

160

Adding Audio Files to the Arrangement

169

Editing Audio Regions

172

Setting a Region's Anchor Point

178

Repeating Audio Regions

181

Working with Sampling Rates

185

Working with Tempo and Audio Regions

190

195

TC: When using the Adjust Tempo using Region Length and Locators command in the Arrange window, Logic will apply the the last used settings the last time the same operation was done in the Sample editor.

Using Folders

199

TC: This section inappropriately uses the term "objects" when it should be "regions". Before Logic 7, Logic referred to the boxes in the Arrange window as objects. The term "objects" is still used in Logic 7, but only to describe items in the environment. Regions is a term that is more consistent with other music production software.

Saving the Song as a Project

203

TS: This section is just like the process done on page 101, but instead of creating a project folder when the song is created, the project folder can be created after the song is finished.

206

TS: The lesson steps on this page instruct the reader to save their song as a project after each of the next six chapters. In a group class setting, it is unnecessary to do this. Each lesson has a start file that is a sufficient beginning point for each lesson.

What You've Learned

207

Lesson 5

Working with Apple Loops

209

TS: The intro to this lesson indicates that Apple Loops first appeared in Soundtrack. As many of your students will likely have experience with Garageband, you may want to point out that this is the same Apple Loop system.

Taking a First Look at Apple Loops

210

212

TS: This is the only section in the book that discusses the ability to make your own Apple Loops. Time permitting, this is a worthwhile feature to explore at the end of the lesson. The Advanced Logic Pro 7 book deals with this feature in more detail.

Working with Tracks

213

TS: This exercise seems somewhat out of place considering the lesson title. You may want to introduce this information at an earlier time.

220

TT: Step 3 converts the track into stereo. A major point of confusion for many audio engineers is how to narrow or off center the stereo image. This is a good point to show the Direction Mixer plug-in, found under the Helper category.

221

TC: This page says it deals with Naming Tracks, but actually you are renaming the audio objects. Command clicking on a track is actually used to add or change a track name. The concern since the students have not been taught what audio objects in the environment are, then this could cause confusion and side-track from the main point of the lesson. You may not want to bring this up at this time, but later in Lesson 12 you may want to address this issue.

Looping Objects in the Arrange Window

223

*TC: This heading should be Looping **Regions** in the Arrange Window. Use of the term objects could cause confusion with your students, so indicate to them that this exercise is dealing with the regions that they are already familiar with.*

224

TS: The note at the top of this page about the Song Information window can become a big distraction. It is not necessary to talk about this window during this lesson, so you may want to simply skip it altogether.

Using the Loop Browser

227

231

TT: Some people like to use the Favorites option mark what loops they've used in any of the songs they've composed.

236

TC: When auditioning Apple Loops, they are played through the last available Audio Track, just as Klopfgest is played through the last available Audio Instrument. Avoid modifying the last Audio Track.

Using the Global Tempo Track

243

244

TC: In order for step 1 to work, the Global Tracks feature must be turned on first.

Finishing the Arrangement

249

What You've Learned

250

Lesson 6

Editing MIDI in the Matrix Editor

253

TS: Now that your students are already familiar with editing in the Arrange window, working with the Matrix will be easy for them to understand. For the most part, the basics concepts of working in the Matrix are identical to the Arrange window. As you see similarities, point them out to your students.

Getting Ready to Edit MIDI

254

Exploring the Matrix Editor

257

259

TC: The first paragraph indicates that clicking on the keyboard in the Matrix, plays the note, but this is only true if the MIDI Out button is selected.

TT: Clicking on a note in the keyboard also selects all MIDI events assigned to that note, in a similar way to how clicking on a track in the Arrange window, selects all regions on that track.

261

TS: The Adding Notes with a MIDI Controller section goes through what most other sequencers refer to as step recording. Mention this, as many of your students may already be familiar with this concept.

TT: The step recording approach to entering notes into Logic is not only available in the Matrix editor, but the Score and Event List editors as well.

TT: This is a great place to use the Caps Lock Keyboard.

262

TC: The second from the bottom paragraph is incorrect. If the Step Input Keyboard is not open, then the default duration is derived from the Division box located below the toolbox. Once the Step Input Keyboard is opened then the value in it overrides the Division box.

TT: Step recording is great place to use the Caps Lock Keyboard.

Using HyperDraw

270

TC: Logic 7 Express is more limited in the number of control values that can be edited with HyperDraw.

277

TT: Hyper Draw can also be accomplished in the Arrange window, however the procedures for copying nodes discussed here will not work with Hyper Draw in the Arrange window, which is discussed on page 284.

What You've Learned

284

Lesson 7

Editing MIDI in the Hyper Editor and Event List

287

Exploring the Hyper Editor

288

291

TS: It should be noted that the Crosshair Tool only appears in the Hyper editor, and although it looks very much like the Marquee Tool, it does not function the same way.

292

TC: Step 5 and the Tip on this page are incorrect. Using the Control key is not necessary. Ctrl-clicking will cause zooming of the selected area.

Using Hyper Sets

293

Exploring the Event List Editor

301

TS: Share with students that the Event List is a way of looking at MIDI in the same way Logic does, in a numeric (mostly) form. Event Lists were the first way for people to edit their MIDI data.

What You've Learned

307

Lesson 8

Recording

309

TS: This is the moment that most students have been waiting for... recording! This section deserves a monumental buildup.

Setting Up the Metronome

310

TC: The note indicates that Klopfgest will be already inserted on Audio Instrument 128. This is true for Logic Pro, but in Logic Express, Klopfgest will be inserted on Audio Instrument 64, as that is the maximum number of instruments available.

311

TS: Setting up a click track is so important that you may want to take a moment and further explain the click options. For example, some people are very particular about the type of click sound they want, so using a MIDI click is a helpful feature.

Setting a Count-in

312

TC: The comments in step 8 indicate that nothing will be recorded during the count-in. This is not the case. Anything played in the count-in is recorded.

Recording MIDI

314

Filtering MIDI Input

321

TS: You should explain why MIDI input filtering can be useful. For example, many keyboard controllers generate channel pressure commands, but the sounds you're using do not react to that information. This creates unnecessary data that can start to "clog" MIDI transmissions.

323

TT: MIDI Thru filtering is still non-destructive. Logic remembers the way the part was actually played, even though MIDI Thru changes the values at the output of the track.

Using Capture Last Take (Logic Pro Only)

326

Setting Audio Options

329

TC: Logic Express only supports sample rates up to 96kHz.

Getting Ready to Record

341

Recording

346

Punch-Recording

349

Merging Recordings

354

Cleaning Up After Recording

356

What You've Learned

359

Lesson 9

Mixing

361

TS: Although the lesson title is Mixing, more emphasis could be placed here on the mixing process. After finishing this lesson, this is a great place to let your students experiment on their own!

TS: Relate this section to the idea of being in a studio. If Logic's software capabilities were transformed into a real studio, it would be one phenomenal studio!

TC: The Master Fader object is a component that is not mentioned in the book, but must be explained. Somewhere in this lesson would be a good place to discuss it. The Master Fader object in the environment should most always be at 0dB. If it is lowered, it actually impacts the volume of all output faders. If students use Logic Control, Mackie Control, or some other control surfaces, the physical master fader actually controls the Master Fader object, which can lead to major confusion about gain structure within a song.

Setting Volume Levels

362

Using the Track Mixer

368

372

TS: The use of the Global track button often causes confusion. Explain that when Global is off, then the Track Mixer reflects the Arrange window, but when Global is on, the Track Mixer reflects the Environment.

Exploring the Fader Area

374

Creating Fades

379

Using Sends

383

TT: Unlike most consoles, the busses in Logic are stereo busses. For example, Bus 1 carries a Left and Right channel.

Monitoring System Performance

390

Using the Freeze Function

391

TT: There is no loss in quality when freezing a file. Freeze files are actually 32 bit quality.

What You've Learned

398

Lesson 10

Automating the Mix

401

TS: There are a lot of little nuances in the Lesson that make Holding-Hands a suggested approach.

TS: The end of this lesson signifies the end of the project song that started with Lesson 4. If you skipped the Bouncing section in Lesson 1, then you should go through it at the end of this lesson, using this song as your bounce material.

Using Track Automation

402

TC: The term “objects” in the first paragraph should be “regions”.

405

TS: On step 8, make sure to tell the students that the pointer must be the automation line, otherwise hitting Control will cause the magnifying glass to appear.

Using Live Automation

411

415

TS: This is the only page in the book where Undo History is discussed, so spend a little time covering this.

What You’ve Learned

426

TS: As this is the end of the Creating a Song section of the book, if time permits, this is a great place to give your students to experiment with their new found knowledge. A fun activity is to allocate a certain amount of time to “remix” the song, and then have them share their remix with the rest of the class. Have them discuss what features they found most useful.

Section III

Customizing Your Setup

427

TS: Lessons 11 and 12 are about Logic's Environment. This part of Logic is considered by many to be one of the most complex and confusing areas in Logic. Before moving into these lessons, explain that the Environment serves as the management system for all incoming and outgoing audio and MIDI. An element that causes confusion is that Logic comes with pre-defined environments, that get many Logic users started, but never require people to develop an understanding of what the Environment is for. Everything is fine until there is a problem. If the user doesn't know how Logic's basic input output system works, then they can never expect to find and fix their problems.

TS: Its often useful to use abstract metaphors to present a new concept. Environment means "surroundings". At the heart of Logic you have a program that can listen and talk to other "things" that surround it. These things are your audio interfaces, MIDI interfaces, and the equipment they are hooked up to. Have your students imagine Logic as an entity floating in the universe. The environment represents the universe that Logic exists within, and things (objects) that are present within Logic's universe. The user must create objects for Logic to listen to, and talk to. If there is nothing present in Logic's universe (the environment) then Logic serves no purpose.

TS: Since everything coming in or out Logic goes through the Environment, some people feel its better to introduce the environment earlier in the class. If students already know about the Environment, it can make it easier for them to understand why Logic behaves certain ways in other parts of the program. The downside is that the Environment can potentially be technical downer that turns people off before they get to know the more creative parts of the program. If you consider teaching the Environment earlier, a good place to cover Lessons 11 and 12 would be just before the Creating a Song section of the book that begins at Lesson 4.

Lesson 11

Setting Up the MIDI Environment

429

TS: This lesson is extremely critical to anyone with any amount of outboard MIDI controllable equipment. Take a survey of your class, and find out what kinds of outboard MIDI gear they have.

Getting Started in Logic's Environment Window

430

Activating Logic's Sequencer

443

Creating a Standard Instrument

450

Creating a Multi Instrument

454

Using Instruments

458

TS: This exercise uses the Quicktime Synth. This is a good time to explain that even virtual software "things" within your computer must still be represented as objects in Logics "universe".

What You've Learned

471

Lesson 12

Setting Up the Audio Environment

473

TS: The first part of this lesson deals with configuring your Audio Driver. This is good place to show how Logic integrates with AMS.

Configuring Audio Hardware and Drivers

474

Exploring the Audio Configuration Window (Logic Pro Only)

478

480

TS: This is the first page that the Master object is mentioned in the book. This is a very important object. It is discussed in more detail in the surround sound Lesson 14, page 542. People that do not have surround playback systems often skip that lesson, but knowing what the Master object does cannot be passed over.

Using Audio Objects

482

Exploring Audio Object Parameters

486

Looking at the Channel View

490

TC: The EQ Area indicates that it can reveal an equalizer with spectrum analyzer. The Spectrum Analyzer is only available in Logic Pro.

Finishing the Autoload Song

492

What You've Learned

496

Section IV

Logic for Video Editors

497

*TS: The last two lesson of the book are in a section that indicates its for video editors. As the majority of your students will likely have an interest in Logic for its music applications, there is a tendency to believe that in the interest of time the last two lessons can be dropped. **Do NOT drop lessons 13 and 14.** There is valuable information in both of these lessons that will apply to anyone using Logic for any reason.*

Lesson 13

Working with video in Logic Pro

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Understanding Video Files and Logic

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Using the Global Video Track

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Working with the Movie's Audio

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Working with SMPTE Timecode

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Scoring to Video

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TS: At the bottom of this page is the only time in the book that Drag Modes are mentioned. This is a powerful feature in Logic, and very useful in musical applications. Time permitting you may want to create a small music exercise that incorporates Drag Modes.

TC: The Shuffle Drag Mode only exists in Logic Pro.

Exporting Audio to Movies

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What You've Learned

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Lesson 14

Working with Surround Sound

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TC: The obvious problem occurs in that most classroom environments do not have a surround sound system. It is possible to still work through this lesson even though the students will not be able to hear the results. There are valuable lessons to learn in this lesson even if surround sound is never used.

TS: A potential problem is that the majority of your students will not have much experience with surround sound. This can potentially lead to a lot of questions that have nothing to do with Logic specifically, so be careful that you don't depart from the main goal.

Configuring the Playback System

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Configuring Logic

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Using the Surround Control

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Finishing a Surround Mix

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TS: Although the Master object is first mentioned on page 480, this is the first time that it is shown in any kind of detail. A point that needs to be heavily emphasized is that the Master object fader has the same impact on all audio outputs regardless of if surround sound is being used.

What You've Learned

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